Supplementary Material for When Does Egocentric Human Data Help Robot Policies? A Spectral Theory of Cross-Embodiment Data Mixing

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A. Setup and notation

We restate the model of the main text (Sec. 3) for self-containment. Coordinates $j = 1, \dots, p$ index a shared spectral basis; domain $e \in \{r, h\}$ has covariates $x_e = \Sigma_e^{1/2} u_e$, u_e standard Gaussian, $\Sigma_e = \mathrm{diag}(\lambda_j^e)$ with $\lambda_j^e \asymp j^{-\alpha_e}$, $\lambda_1^e = 1$; we write $a_j = \lambda_j^r$, $b_j = \lambda_j^h$. Labels: $y_r = \langle \beta^*, x_r \rangle + \varepsilon_r$, $y_h = \langle \beta^* + \delta, x_h \rangle + \varepsilon_h$; $\sum_j j^{2s} \beta_j^{*2} \le 1$; noise variances σ_e^2. Given n_r robot and n_h human samples, mixing ratio $\lambda \in [0, 1]$, per-sample weights $w_r = (1-\lambda)/n_r$, $w_h = \lambda/n_h$, the diagonalized ridge estimator is

$$\hat{\beta}_j = \frac{w_r T_{rj} + w_h T_{hj}}{w_r S_{rj} + w_h S_{hj} + \mu}, \qquad S_{ejk} = \sum_{i=1}^{n_e} x_{e,ij} x_{e,ik}, \quad T_{ej} = \sum_i x_{e,ij}\, y_{e,i}. \tag{1}$$

The robot-domain excess risk is $\mathcal{R}_T(\lambda) = \mathbb{E}\|\hat{\beta}(\lambda) - \beta^*\|_{\Sigma_r}^2 = \sum_j a_j\, \mathbb{E}(\hat{\beta}_j - \beta_j^*)^2$. The effective noise is $\tilde{\sigma}_e^2 = \sigma_e^2 + F_e$ with feedthrough energy $F_e = \sum_k \lambda_k^e \beta_{e,k}^2$ ($\beta_h = \beta^* + \delta$).

Single-coordinate notation (used throughout, index j suppressed). $U_r \sim \chi^2_{n_r}$, $U_h \sim \chi^2_{n_h}$ independent; $S_e = \lambda^e U_e$, so $\mathbb{E}S_e = n_e \lambda^e$, $\mathrm{Var}(S_e) = 2n_e(\lambda^e)^2$, and the third central moment is $8n_e(\lambda^e)^3$. $\hat{D} = w_r S_r + w_h S_h + \mu$; $D = \mathbb{E}\hat{D} = (1-\lambda)a + \lambda b + \mu$. $\hat{N} = w_r T_r + w_h T_h$; its systematic part $N_0 = \lambda b\, \delta - \mu \beta^*$. The fluctuation scale is $\eta = \sqrt{8 \log(4p/\delta_c)}\, \big(\frac{(1-\lambda)a}{\sqrt{n_r}} + \frac{\lambda b}{\sqrt{n_h}} \big)$ for confidence parameter $\delta_c \in (0, 1)$. We write $\tilde{n} = \hat{N} - \mathbb{E}\hat{N}$; direct expansion gives

$$\mathrm{Var}(\tilde{n}) = \underbrace{\frac{2\lambda^2 (b\delta)^2}{n_h}}_{\text{mismatch fluctuation}} + \underbrace{\frac{(1-\lambda)^2 \tilde{\sigma}_r^2 a}{n_r} + \frac{\lambda^2 \tilde{\sigma}_h^2 b}{n_h}}_{\text{noise + feedthrough}}. \tag{2}$$

B. Lemmas L1 and L2: per-coordinate representation and conditional MSE

Lemma 1 (L1, representation)**.** *For each coordinate j, with the definitions above,*

$$T_{ej} = S_{ejj}\, \beta_{e,j} + U_{ej} + \textstyle\sum_i x_{e,ij} \varepsilon_{e,i}, \qquad U_{ej} = \textstyle\sum_{k \ne j} S_{ejk}\, \beta_{e,k}, \tag{3}$$

where U_{ej} is mean-zero with $\mathrm{Var}(U_{ej}) = n_e \lambda_j^e F_{e,j}$, $F_{e,j} = \sum_{k \ne j} \lambda_k^e \beta_{e,k}^2$.

Proof. Expand $y_{e,i} = \sum_k x_{e,ik} \beta_{e,k} + \varepsilon_{e,i}$ in $T_{ej} = \sum_i x_{e,ij} y_{e,i}$ and separate the $k = j$ term. For $k \ne j$, $\mathbb{E}S_{ejk} = \sum_i \mathbb{E}[x_{e,ij}]\mathbb{E}[x_{e,ik}] = 0$ by independence of distinct coordinates of the Gaussian covariate, hence $\mathbb{E}U_{ej} = 0$. For the variance, $\mathrm{Var}(S_{ejk}) = \sum_i \mathbb{E}[x_{e,ij}^2]\mathbb{E}[x_{e,ik}^2] = n_e \lambda_j^e \lambda_k^e$ (independence of coordinates $j \ne k$), and distinct cross terms are uncorrelated: for $k \ne l$, $\mathrm{Cov}(S_{ejk}, S_{ejl}) = \sum_i \mathbb{E}[x_{e,ij}^2 x_{e,ik} x_{e,il}] = 0$ since $\mathbb{E}[x_{e,ik} x_{e,il}] = 0$ and $x_{e,ij}^2$ is independent of $(x_{e,ik}, x_{e,il})$. Summing, $\mathrm{Var}(U_{ej}) = \sum_{k \ne j} \beta_{e,k}^2\, n_e \lambda_j^e \lambda_k^e = n_e \lambda_j^e F_{e,j}$. Finally, $F_{e,j} = F_e - \lambda_j^e \beta_{e,j}^2 = F_e(1 + o(1))$ under the ellipsoid constraint, since the omitted single term is negligible against the total energy. □

Remark 1. The feedthrough U_{ej} is indistinguishable from label noise for coordinate j: it contributes variance $n_e \lambda_j^e F_e$ exactly as if the noise variance were $\sigma_e^2 + F_e$. This is the origin of the effective noise $\tilde{\sigma}_e^2$ in Eq. (2); absorbing it there is justified because U_{ej} is independent of S_{ejj} (disjoint coordinate sets) and of the label noise.

Lemma 2 (L2, conditional MSE)**.** *Conditioned on the design matrices,*

$$\mathbb{E}\big[(\hat{\beta}_j - \beta_j^*)^2 \,\big|\, X\big] = \frac{\hat{N}_j^2}{\hat{D}_j^2},$$
$$\hat{N}_j = w_h S_{hjj} \delta_j - \mu \beta_j^* + w_r U_{rj} + w_h U_{hj} + w_r X_{r,:j}^\top \varepsilon_r + w_h X_{h,:j}^\top \varepsilon_h. \tag{4}$$

Proof. Substitute Lemma 1 into Eq. (1): $\hat{\beta}_j \hat{D}_j = w_r S_{rjj}\beta_j^* + w_h S_{hjj}(\beta_j^* + \delta_j) + (\text{feedthrough} + \text{noise})$. Subtract $\beta_j^* \hat{D}_j = w_r S_{rjj}\beta_j^* + w_h S_{hjj}\beta_j^* + \mu\beta_j^*$; the error numerator is exactly $\hat{N}_j$. Conditional on X, the only randomness left is the label noise, which is mean-zero; the displayed expression is the conditional second moment, with the noise contributions' conditional variances $w_e^2\sigma_e^2 S_{ejj}$ understood as part of $\hat{N}_j^2$ after taking expectations over the noise (cross terms vanish by independence and mean-zero). □

C. Lemma L3: per-coordinate deterministic equivalent

Lemma 3 (L3). *Fix a coordinate and suppose $D \geq 2\eta$ (information-rich regime). Define*

$$\mathrm{MSE}^{\mathrm{det}} = \frac{N_0^2}{D^2} + \frac{\mathrm{Var}(\tilde{n})}{D^2} - \frac{4N_0\,\mathrm{Cov}(\tilde{n},\hat{D})}{D^3} + \frac{3N_0^2\,\mathrm{Var}(\hat{D})}{D^4}, \tag{5}$$

with $\mathrm{Var}(\tilde{n})$ from Eq. (2), $\mathrm{Cov}(\tilde{n},\hat{D}) = 2\lambda^2\delta b^2/n_h$, and $\mathrm{Var}(\hat{D}) = 2\big[\frac{(1-\lambda)^2a^2}{n_r} + \frac{\lambda^2 b^2}{n_h}\big]$. Then

$$\Big|\,\mathbb{E}\big[(\hat{\beta}-\beta^*)^2\big] - \mathrm{MSE}^{\mathrm{det}}\,\Big| \;\leq\; C\Big(\frac{\eta}{D}\Big)^3\frac{\overline{N}^2}{D^2} + O\big(\delta_c\,\mu^{-2}\,\overline{N}^2\big), \tag{6}$$

where $\overline{N}^2 = N_0^2 + \mathrm{Var}(\tilde{n})$ and C is an absolute constant.

Before the proof we record that only the mismatch fluctuation correlates with $\hat{D}$: $\mathrm{Cov}(\tilde{n},\hat{D}) = \mathrm{Cov}(w_h\lambda^h(U_h - n_h)\,\delta,\; w_h\lambda^h U_h) = w_h^2(\lambda^h)^2\delta\,\mathrm{Var}(U_h) = 2\lambda^2\delta b^2/n_h$, since the noise and feedthrough parts of $\tilde{n}$ are independent of (U_r, U_h).

Proof. Step 1 (concentration of $\hat{D}$). Write $\hat{D} - D = w_r\lambda^r(U_r - n_r) + w_h\lambda^h(U_h - n_h)$. By the Laurent–Massart χ^2 tail bound, $\Pr(|U_e - n_e| \geq 2\sqrt{n_e x} + 2x) \leq 2e^{-x}$; taking $x = \log(4p/\delta_c)$ and a union bound over the p coordinates, with probability $\geq 1 - \delta_c$ simultaneously for all coordinates $|\hat{D} - D| \leq \eta$. On the complementary event, $|\hat{N}^2/\hat{D}^2| \leq \hat{N}^2\mu^{-2}$, and $\hat{N}$ has finite fourth moments (those of a χ^2 mixture plus Gaussian quadratics), so the bad event contributes $O(\delta_c\mu^{-2}\overline{N}^2)$.

Step 2 (Taylor expansion on the good event). Let $f(x) = x^{-2}$. For $|\Delta| \leq \eta \leq D/2$, $f(D+\Delta) = D^{-2} - 2\Delta D^{-3} + R$ with $|R| \leq \frac{1}{2}\max_{\xi\in[D-\eta,D+\eta]} 6\xi^{-4}\Delta^2 \leq 48\,\eta^2/D^4$.

Step 3 (moment expansion). On the good event (Step 1 justifies the truncation of the expansion; the indicator algebra contributes only lower-order terms absorbed into the remainder),

$$\mathbb{E}\big[\hat{N}^2 f(\hat{D})\big] = \frac{\mathbb{E}\hat{N}^2}{D^2} - \frac{2\,\mathbb{E}[\hat{N}^2\Delta]}{D^3} + \mathbb{E}\big[\hat{N}^2 R\big], \qquad \mathbb{E}\hat{N}^2 = N_0^2 + \mathrm{Var}(\tilde{n}). \tag{7}$$

Expanding $\mathbb{E}[\hat{N}^2\Delta] = N_0^2\,\mathbb{E}\Delta + 2N_0\,\mathrm{Cov}(\tilde{n},\hat{D}) + \mathbb{E}[\tilde{n}^2\Delta]$, the first term vanishes since $\mathbb{E}\Delta = 0$. For the third term, decompose $\tilde{n} = a_1 v_h + \zeta$ with $a_1 = \frac{\lambda}{n_h} b\,\delta$, $v_h = U_h - n_h$, and ζ collecting the noise, feedthrough, and robot-side parts, independent of (v_r, v_h) with $\mathbb{E}\zeta = 0$; and $\Delta = w_r\lambda^r v_r + w_h\lambda^h v_h$. Then

$$\begin{aligned}\mathbb{E}[\tilde{n}^2\Delta] &= \mathbb{E}\big[(a_1 v_h + \zeta)^2(w_r\lambda^r v_r + w_h\lambda^h v_h)\big] \\ &= a_1^2 w_h\lambda^h\,\mathbb{E}[v_h^3] = a_1^2\,w_h\lambda^h\cdot 8n_h = \frac{8\lambda^3 b^3\delta^2}{n_h^2},\end{aligned} \tag{8}$$

where all cross terms vanish by independence and $\mathbb{E}v = \mathbb{E}\zeta = 0$, using $\mathbb{E}[v_h^3] = 8n_h$ (third central moment of $\chi^2_{n_h}$). This term contributes $-2\mathbb{E}[\tilde{n}^2\Delta]/D^3 < 0$ to the MSE, of relative order $O(\lambda^3\delta^2/(n_h^2 D^3))$ against the leading terms — analytically exact, numerically negligible; we keep it in the formula for exactness. The remainder satisfies $|\mathbb{E}[\hat{N}^2 R]| \leq 48\eta^2 D^{-4}\,\mathbb{E}\hat{N}^2 = O\big((\eta/D)^2\cdot\overline{N}^2/D^2\big)$.

Step 4 (collection). Substituting Step 3 into Eq. (7) and grouping terms gives Eq. (5) with the announced error: the dominant remainder is $(\eta/D)^3\overline{N}^2/D^2$ (the $\mathbb{E}[\tilde{n}^2\Delta]$ term beyond its exact value, and higher-order Taylor terms). Since $\eta/D = O(n_{\min}^{-1/2})$ up to spectral-weight factors, the relative error is $O(n_{\min}^{-3/2})$, matching the Monte-Carlo accuracy of 1–2% (Sec. H). □

D. Lemma L4: truncation assembly

Lemma 4 (L4). *Let $m = \max\{j : D_j \geq \kappa_0\eta_j\}$ for a fixed threshold $\kappa_0 \geq 2$. Then: (i) on the truncated coordinates $j > m$, $\mathbb{E}(\hat{\beta}_j - \beta_j^*)^2 = \Theta(\beta_j^{*2})$; (ii) on the retained coordinates $j \leq m$, Lemma 3 applies with aggregate relative error $O(\kappa_0^{-3}) + o(1)$; (iii) consequently*

$$\mathcal{R}_T(\lambda) = \sum_{j>m} a_j\beta_j^{*2} + \Big(\sum_{j\leq m} a_j\,\mathrm{MSE}_j^{\mathrm{det}}\Big)\big(1 + O(\kappa_0^{-3} + n_{\min}^{-3/2})\big), \tag{9}$$

and $\sum_{j>m} a_j\beta_j^{*2} \asymp m^{-(\alpha_r+2s)}$.

Proof. (i) For $j > m$, $D_j < \kappa_0\eta_j$; since η_j is the fluctuation scale of the information terms, the information part of D_j is at most of the order of η_j itself, while μ is constant — tracing the definitions gives $D_j < 2\mu$ once the spectral weights have decayed below the ridge floor, hence $\mu/D_j > 1/2$. Then $\mathbb{E}\hat{\beta}_j = \beta_j^*(1 - \mu/D_j) + O(\eta_j/D_j)$ with the shrinkage factor bounded away from 1: the squared bias is $\Theta(\beta_j^{*2})$, and the variance is subleading because $|\hat{\beta}_j| \leq |\hat{N}_j|/\mu$ while $N_0 = -\mu\beta_j^*$ cancels the leading term. The total truncated contribution is $\sum_{j>m} a_j\beta_j^{*2} \asymp m^{-(\alpha_r+2s)}$ by power-law tail integration with $a_j \asymp j^{-\alpha_r}$, $\beta_j^{*2} \asymp j^{-(2s+1)}$.

(ii) Split the retained set into the *well-conditioned core* $\{j : D_j \geq \kappa_0\eta_j\}$ with κ_0 large, where Lemma 3 gives

uniform relative error $O(\kappa_0^{-3})$, and the *boundary band* $\{2\eta_j \le D_j < \kappa_0\eta_j\}$. Under the power-law spectra, the band width solves $D_j = \kappa_0\eta_j$, whose solution set has cardinality $o(m)$; coordinates in the band have risk between the leading term and the truncated value, so their total contribution is absorbed by the constants of the truncation term without changing its order. The aggregate relative error over $j \le m$ is therefore $O(\kappa_0^{-3}) + o(1)$.

(iii) Sum (i) and (ii); infimizing over m yields the theorem statement of the main text (the infimum is attained at the self-consistent truncation by construction). □

E. Proof of Corollary 1 (marginal-benefit decomposition)

Differentiate Eq. (5) at $\lambda = 0^+$ per coordinate. With $D_0 = a + \mu$, $V = \tilde{\sigma}_r^2 a/n_r$, direct expansion gives

$$\partial_\lambda \mathrm{MSE}^{\mathrm{det}}\big|_{0^+} = \frac{-2\mu\beta^* b\,\delta - 2V}{D_0^2} - \frac{2\big(V + \mu^2\beta^{*2}\big)(b-a)}{D_0^3}, \tag{10}$$

which, after flipping the sign and dropping the positive factor $2/D_0^3$, is the three-term decomposition of the main text:

$$\mathrm{benefit}_j \;\propto\; \underbrace{\mu\beta_j^* b_j\delta_j}_{\text{A: alignment}} + \underbrace{V_j D_{0j}}_{\text{B: variance}} + \underbrace{\big(V_j + \mu^2\beta_j^{*2}\big)(b_j - a_j)}_{\text{C: coverage}}. \tag{11}$$

Analytic derivative vs. finite differences on the formula: relative error $< 1\%$ (configuration of the v2 benchmark). At $\mu = 0$, $D_0 = a$, terms A and the shrinkage part of C vanish, and $\mathcal{R}_T'(0^+)|_{\mu=0} = -(2\tilde{\sigma}_r^2/n_r)\sum_j j^{\alpha_r - \alpha_h} < 0$ always.

F. Theorem 2: derivation and exactness of the H-criterion

Take $n_r \to \infty$ with n_h/n_r fixed in Eq. (10): the variance term B vanishes, and for anti-aligned mismatch ($c < 0$; write $|\delta|$ for the effective sign-flipped components) the marginal benefit converges to $-H$ with

$$H = \underbrace{2\mu\sum_j \frac{a_j\beta_j^* b_j|\delta_j|}{D_{0j}^2}}_{A_\infty} \underbrace{- 2\mu^2\sum_j \frac{a_j\beta_j^{*2}(b_j - a_j)}{D_{0j}^3}}_{C_\infty}. \tag{12}$$

At finite n_r the variance terms contribute $-K/n_r$ to the derivative, with $K = 2\tilde{\sigma}_r^2(S_1 + S_2)$, $S_1 = \sum_j a_j^2/D_{0j}^2$, $S_2 = \sum_j a_j^2(b_j - a_j)/D_{0j}^3$; hence marginal mixing is beneficial iff $K/n_r > H$, giving $n_r^{\mathrm{crit}} = K/H$ when $H > 0$. Numerically (v11, $p = 1000$, $\mu = 0.05$, spectral family A): A_∞ is linear in $\|\delta\|$ with slope 6.975×10^{-3}, $C_\infty = 4.700 \times 10^{-3}$ independent of δ, predicting $\|\delta\|^* = 0.674$, which brackets the Monte-Carlo transition exactly; and $n_r^{\mathrm{crit}} \cdot H = 25.866$ is constant across $\|\delta\| \in \{0.75, 1.0, 2.0\}$ to display precision.

The classical limit. As $\mu \to 0$, $C_\infty \to 0$ and $H = A_\infty > 0$ whenever $c < 0$: the discard-the-source transition of classical domain adaptation is recovered. At $\mu > 0$ the large-n_r mixing benefit tends to $-C_\infty < 0$: regularization carries a nonzero coverage benefit even with unlimited robot data.

G. Corollary 2 (exchange rate): existence, uniqueness, and unimodality

κ_{ex} is defined implicitly by $\bar{\mathcal{R}}_T(\lambda^*; n_r, n_h) = \bar{\mathcal{R}}_T(0;\, n_r + \kappa_{\mathrm{ex}} n_h)$, where λ^* minimizes the mixture risk over $[0,1]$. This section proves that κ_{ex} is well-defined without any unimodality assumption (Proposition 5), that every per-coordinate risk is unimodal with an explicit minimizer (Lemma 6 and Proposition 7), and that global unimodality holds exactly in the matched-spectrum regime (Theorem 8) and under an explicit convexity condition in general (Proposition 9).

G.1. Well-definedness of the exchange rate

Proposition 5 (existence and uniqueness of κ_{ex})**.** *Fix (n_r, n_h) and write $m_\star = \min_{\lambda\in[0,1]} \bar{\mathcal{R}}_T(\lambda; n_r, n_h)$ (attained: $\bar{\mathcal{R}}_T$ is continuous in λ on a compact set). As a function of the robot sample size n at $\lambda = 0$,*

$$\bar{\mathcal{R}}_T(0; n) \;=\; \sum_{j>m} a_j\beta_j^{*2} \;+\; \sum_{j\le m} a_j\,\frac{\tilde{\sigma}_r^2 a_j/n + \mu^2\beta_j^{*2}}{(a_j+\mu)^2} \tag{13}$$

is continuous and strictly decreasing in n whenever $\tilde{\sigma}_r^2 > 0$ (in particular whenever $\beta^* \neq 0$, since $\tilde{\sigma}_r^2 = \sigma_r^2 + F_r > 0$), with limit

$$\bar{\mathcal{R}}_T(0;\infty) \;=\; \sum_j a_j\,\frac{\mu^2\beta_j^{*2}}{(a_j+\mu)^2}, \tag{14}$$

the pure shrinkage risk (the truncation recedes, $m \to p$, as $n \to \infty$). Since $\lambda = 0$ is feasible, $m_\star \le \bar{\mathcal{R}}_T(0; n_r)$, and therefore: (i) if $m_\star > \bar{\mathcal{R}}_T(0;\infty)$, there exists a unique $\kappa_{\mathrm{ex}} \ge 0$ solving the defining equation; (ii) if $m_\star \le \bar{\mathcal{R}}_T(0;\infty)$, no finite amount of additional robot data reproduces the mixture optimum — this is the cap regime of Corollary 2(iv) of the main text. Only the minimum value *$m_\star$ enters the construction; unimodality of $\bar{\mathcal{R}}_T(\cdot)$ is not used.*

Proof. On the retained coordinates, $\partial_n \bar{\mathcal{R}}_T(0;n) = -\sum_{j\le m} a_j^2\,\tilde{\sigma}_r^2/\big(n^2(a_j+\mu)^2\big) < 0$ (the truncation index m is locally constant in n almost everywhere), and $\bar{\mathcal{R}}_T(0;n) \downarrow \bar{\mathcal{R}}_T(0;\infty)$ as $n \to \infty$. The map $\kappa \mapsto \bar{\mathcal{R}}_T(0; n_r + \kappa n_h)$ is thus a continuous strict bijection from $[0,\infty)$ onto $(\bar{\mathcal{R}}_T(0;\infty),\, \bar{\mathcal{R}}_T(0;n_r)]$, which contains $m_\star$ in case (i) and not in case (ii). □

G.2. Per-coordinate unimodality, exactly

Drop the coordinate index and write a per-coordinate risk (effective variance plus bias square, including the mismatch-fluctuation term of Eq. (2)) as $\mathrm{MSE}(\lambda) = N(\lambda)/D(\lambda)^2$ with

$$N(\lambda) = v_r(1-\lambda)^2 + v_h\lambda^2 + (u\lambda - w)^2 + \frac{2u^2}{n_h}\lambda^2, \quad (15)$$

and $D(\lambda) = D_0 + \Delta\lambda$, where $v_e = \tilde{\sigma}_e^2\lambda^e/n_e$ are the variance prices, $u = b\delta$, $w = \mu\beta^*$, $D_0 = a+\mu > 0$, $\Delta = b-a$, and $D > 0$ on $[0,1]$ since $D(1) = b+\mu > 0$. N is a quadratic, $N(\lambda) = n_0 + n_1\lambda + n_2\lambda^2$, with $n_0 = v_r + w^2$, $n_1 = -2(v_r + uw)$, $n_2 = v_r + v_h + u^2(1 + 2/n_h)$.

Lemma 6 (affine derivative numerator)**.** *$\partial_\lambda \mathrm{MSE}(\lambda) = L(\lambda)/D(\lambda)^3$ with L affine:*

$$L(\lambda) = p + q\lambda, \qquad p = n_1 D_0 - 2\Delta n_0, \qquad q = 2n_2 D_0 - \Delta n_1. \quad (16)$$

Proof. $\partial_\lambda \mathrm{MSE} = (N'D - 2\Delta N)/D^3$; in $N'D - 2\Delta N$ the quadratic terms $2n_2\Delta\lambda^2$ cancel exactly. □

Proposition 7 (coordinate risks are unimodal)**.** *L cannot cross zero from above on $[0,1]$: the pattern $L(0) > 0 > L(1)$ never occurs. Hence each coordinate risk is quasiconvex on $[0,1]$, with explicit minimizer*

$$\lambda^* = \begin{cases} 0, & L(0) \geq 0, \\ -p/q \in (0,1), & L(0) < 0 < L(1), \\ 1, & L(1) \leq 0. \end{cases} \quad (17)$$

Proof. With $D_1 = D_0 + \Delta = b + \mu > 0$ and $s = uD_0 + \Delta w$, direct rearrangement gives the endpoint values

$$\begin{aligned} L(0) &= -2\big(v_r D_1 + ws\big), \\ L(1) &= 2\Big(\big(v_h + \tfrac{2u^2}{n_h}\big)D_0 + (u-w)\,s\Big). \end{aligned} \quad (18)$$

Suppose $L(0) > 0 > L(1)$. Since $v_r \geq 0$, $v_h + 2u^2/n_h \geq 0$, and $D_0, D_1 > 0$: if $s \geq 0$, the first inequality forces $w < 0$ and the second forces $u < w$; hence $s = uD_0 + \Delta w < w(D_0 + \Delta) = wD_1 < 0$, contradicting $s \geq 0$. If instead $s < 0$, they force $w > 0$ and $u > w$; hence $s > wD_1 > 0$, contradicting $s < 0$. The remaining sign patterns of the affine L give the classification (17); an interior zero of L is a sign change from negative to positive, hence a minimum. □

Remark 2 (sanity checks). At $\delta = 0$ and $\Delta = 0$ the interior branch of (17) reduces to $\lambda^* = v_r/(v_r + v_h)$, the classical inverse-variance weight. Moreover $L(0) = D_0^3\,\partial_\lambda \mathrm{MSE}(0^+)$, so the coordinate-wise marginal-harm condition $L(0) \geq 0$ is exactly the three-term decomposition of Corollary 1 of the main text.

G.3. Global unimodality: an exact region and a sufficient condition

Sums of quasiconvex functions need not be quasiconvex, so the aggregate $\bar{\mathcal{R}}_T(\lambda) = \sum_j a_j\,\mathrm{MSE}_j(\lambda)$ requires a separate argument.

Theorem 8 (matched spectra: quadratic regime)**.** *If $b_j = a_j$ for all j, then $\bar{\mathcal{R}}_T$ is a quadratic polynomial on $[0,1]$, $\bar{\mathcal{R}}_T(\lambda) = A_2\lambda^2 + A_1\lambda + A_0$, with*

$$A_2 = \sum_j \frac{a_j\, n_{2j}}{(a_j+\mu)^2}, \qquad A_1 = \sum_j \frac{a_j\, n_{1j}}{(a_j+\mu)^2}. \quad (19)$$

Whenever $A_2 > 0$ — i.e., unless the model is degenerate ($v_{rj} = v_{hj} = u_j = 0$ for every j) — $\bar{\mathcal{R}}_T$ is strictly convex, with the unique closed-form minimizer

$$\lambda^* = \mathrm{clip}_{[0,1]}\left(\frac{\sum_j a_j(v_{rj} + u_j w_j)/(a_j+\mu)^2}{\sum_j a_j\, n_{2j}/(a_j+\mu)^2}\right). \quad (20)$$

Proof. With $\Delta_j = 0$, $\mathrm{MSE}_j(\lambda) = N_j(\lambda)/(a_j+\mu)^2$; summing with weights a_j gives the quadratic, and the interior minimizer is $-A_1/(2A_2)$. □

Proposition 9 (convexity region)**.** *The per-coordinate second derivative is*

$$\mathrm{MSE}''(\lambda) = \frac{N''(\lambda)D(\lambda)^2 - 4\Delta\, N'(\lambda)D(\lambda) + 6\Delta^2 N(\lambda)}{D(\lambda)^4}, \quad (21)$$

and a sufficient condition for convexity of MSE_j on $[0,1]$ — hence, summing over j, for strict convexity and unimodality of $\bar{\mathcal{R}}_T$ — is

$$2\,|\Delta_j|\,\big(|n_{1j}| + 2n_{2j}\big)\,D_j^{\max} \;\leq\; n_{2j}\,\big(D_j^{\min}\big)^2 \qquad \text{for every } j, \quad (22)$$

where $D_j^{\max} = \max(a_j, b_j) + \mu$ and $D_j^{\min} = \min(a_j, b_j) + \mu$. Since $n_{2j} > 0$ and $D_j^{\min} > 0$, the condition holds whenever the per-coordinate spectral mismatch $|\Delta_j|/(a_j+\mu)$ is small enough, and is automatic at zero spectral gap.

Proof. Differentiate Lemma 6: $\mathrm{MSE}'' = (qD - 3\Delta L)/D^4$; substituting $q = N''D - \Delta N'$ and $L = N'D - 2\Delta N$ gives the displayed numerator. Since N is quadratic, $N'' = 2n_2$, $|N'| \leq |n_1| + 2n_2$ on $[0,1]$, $D \in [D^{\min}, D^{\max}]$, and $6\Delta^2 N \geq 0$; the numerator is therefore at least $2n_2(D^{\min})^2 - 4|\Delta|(|n_1| + 2n_2)D^{\max}$, which is nonnegative under (22). □

Outside the region (22) we have no proof of global unimodality, and positive combinations of quasiconvex rationals with distinct denominators can in principle be multimodal; however no multimodal instance was found in a randomized search over 4×10^3 aggregates (power-law and adversarial spectra), and $\bar{\mathcal{R}}_T(\lambda)$ is numerically unimodal on all grids used in this paper (Secs. F and H). In any case, Proposition 5 guarantees that κ_{ex} is well-defined irrespective of unimodality.

G.4. The source-sample axis: interior minimizers and an exact dosage law

Corollary 10 (interior minimizers from endpoint signs)**.** *If* $\bar{\mathcal{R}}'_T(0^+) = \sum_j a_j L_j(0)/D_{0j}^3 < 0$ *(the condition of Corollary 1 of the main text) and* $\bar{\mathcal{R}}'_T(1^-) = \sum_j a_j L_j(1)/D_{1j}^3 > 0$*, with* $L_j(0), L_j(1)$ *from Eq. (18) and* $D_{1j} = b_j + \mu$*, then neither* $\lambda = 0$ *nor* $\lambda = 1$ *is optimal and every global minimizer lies in* $(0, 1)$*. No unimodality is needed for existence.*

Proof. $\bar{\mathcal{R}}'_T(0^+) < 0$ yields points with risk strictly below $\bar{\mathcal{R}}_T(0)$, and $\bar{\mathcal{R}}'_T(1^-) > 0$ yields points strictly below $\bar{\mathcal{R}}_T(1)$; a global minimizer exists by continuity on the compact interval. □

Lemma 11 (monotone value in n_h)**.** *At fixed* λ*, each coordinate risk is nonincreasing in* n_h*: both human-sample terms of* N*,* $v_h\lambda^2 = \tilde{\sigma}_h^2 b\lambda^2/n_h$ *and* $2u^2\lambda^2/n_h$*, have nonnegative coefficients. Hence the optimal gain* $g(n_h) = \bar{\mathcal{R}}_T(0) - \min_\lambda \bar{\mathcal{R}}_T(\lambda; n_h)$ *is nondecreasing: more human data never hurts at the optimal mixing ratio.*

Theorem 12 (dosage law; matched spectra)**.** *Assume* $b_j = a_j$ *for all* j *and* $A_1 = \bar{\mathcal{R}}'_T(0^+) < 0$*, and write* $A_2(n_h) = A_2^r + B/n_h$ *with*

$$A_2^r = \sum_j \frac{a_j(v_{rj} + u_j^2)}{(a_j + \mu)^2}, \qquad B = \sum_j \frac{a_j(\tilde{\sigma}_h^2 b_j + 2u_j^2)}{(a_j + \mu)^2}. \tag{23}$$

On the range where the minimizer is interior $(-A_1 < 2A_2(n_h))$*:*

(i) the optimal gain is hyperbolic, $g(n_h) = A_1^2/\big(4A_2(n_h)\big)$*, with* $g(0^+) = 0$*,* $g(\infty) = A_1^2/(4A_2^r)$*, and half-saturation point* $n_h^{1/2} = B/A_2^r$*;*

(ii) the worth-mixing threshold at tolerance $\varepsilon < g(\infty)$ *is*

$$g(n_h) \geq \varepsilon \iff n_h \geq n_h^\varepsilon = \frac{4\varepsilon B}{A_1^2 - 4\varepsilon A_2^r}; \tag{24}$$

(iii) the optimal ratio grows smoothly from zero, $\lambda^*(n_h) = -A_1/\big(2A_2(n_h)\big) \sim (-A_1/2B) \cdot n_h$ *as* $n_h \to 0$*, approaching* $\lambda_\infty^* = -A_1/(2A_2^r)$*;*

(iv) the exchange rate has the closed form

$$\kappa_{\mathrm{ex}}(n_h) = \frac{1}{n_h} \cdot \frac{n_r^2\, g(n_h)}{C - n_r\, g(n_h)}, \qquad C = \tilde{\sigma}_r^2 \sum_j \frac{a_j^2}{(a_j + \mu)^2}, \tag{25}$$

since at matched spectra $\bar{\mathcal{R}}_T(0; n) = C/n + \mathrm{const}$*; the cap regime of Proposition 5(ii) is the pole* $g(n_h) \geq C/n_r$ *(*$\kappa_{\mathrm{ex}} = +\infty$*). Note* $C = K/2$ *at zero spectral gap, with* K *the constant of Theorem 2 of the main text.*

Proof. By Theorem 8, $\bar{\mathcal{R}}_T(\lambda; n_h) = A_2(n_h)\lambda^2 + A_1\lambda + A_0$, and only A_2 depends on n_h. An interior minimizer gives $g = A_1^2/(4A_2)$ and $\lambda^* = -A_1/(2A_2)$; (i)–(iii) are rearrangements, with λ^* nondecreasing in n_h since A_2 decreases. For (iv), $\bar{\mathcal{R}}_T(0; n) = C/n + \sum_j a_j\mu^2\beta_j^{*2}/(a_j + \mu)^2$; solving $C/(n_r + \kappa_{\mathrm{ex}} n_h) = C/n_r - g(n_h)$ for κ_{ex} yields the formula, whose denominator vanishes exactly at $g(n_h) = C/n_r$ — the cap condition $m_\star \leq \bar{\mathcal{R}}_T(0; \infty)$ of Proposition 5(ii). □

If $-A_1 \geq 2A_2(n_h)$ the minimizer sits at the boundary $\lambda^* = 1$ and $g(n_h) = -A_1 - A_2(n_h)$; since $A_2(n_h)$ decreases in n_h, the crossing happens at most once, at $n_h^1 = 2B/(-A_1 - 2A_2^r)$.

Remark 3 (the second transition is of a different kind). The two axes of the phase diagram differ in kind. Along n_r there is a genuine transition (Theorem 2 of the main text): the sign of $\bar{\mathcal{R}}'_T(0^+)$ flips at $H = 0$ and beyond n_r^{crit} all human data should be discarded. Along n_h there is no sign change: once $\bar{\mathcal{R}}'_T(0^+) < 0$, human data helps at *every* sample size, and the classical small-source threshold softens into the explicit n_h^ε of (ii) — equivalently, $\lambda^*(n_h) \leq \lambda_0$ for all $n_h \leq 2B\lambda_0/(-A_1 - 2A_2^r\lambda_0)$. The hard classical switch is recovered as the divergence $n_h^\varepsilon \to \infty$ when $A_1 \to 0^-$ or $B \to \infty$ (noisy or badly mismatched human data).

The three monotonicities of Corollary 2 are read off term by term from Eq. (11): increasing $\alpha_r - \alpha_h$ enlarges the coverage term C on high-risk coordinates; increasing the mismatch enlarges A-harm; increasing n_r shrinks B. The cap regime (case (ii) of Proposition 5) appears when the human spectrum dominates on coordinates the robot spectrum has abandoned: the right-hand side of the defining equation is then bounded below by the truncation tail $\sum_{j>m'} a_j\beta_j^{*2}$ for any robot-only sample size, while the mixture side eliminates part of that tail through the human samples. Numerics (v12): $n_r = 1000$, $n_h/n_r = 10$, $\mu = 0.05$, white aligned mismatch $\|\delta\| = 0.3$, $\sigma_h = 2\sigma_r$; per-α_h Spearman correlations between κ_{ex} and the spectral gap are 0.55–0.82; the cap binds from $\alpha_r - \alpha_h \approx 0.5$–$0.75$; at zero gap $\kappa_{\mathrm{ex}} \in [0.42, 0.59]$.

H. Numerical verification protocol

(v5) Accuracy of the deterministic equivalent. Three spectral/mismatch configurations × two mixing ratios, 200 Monte-Carlo replicates (∼200k coordinate-replicates): binned median relative error 0.6–1.6%, aggregate and Σ_r-weighted aggregate error $\leq 2\%$. Ablations v3/v4 show the feedthrough term — not the ratio-fluctuation corrections — is the dominant closure: zero-order (population) formulas underestimate by 30–90%, ratio corrections alone do not close the gap, and feedthrough closes it to the stated accuracy.

(v6) Full ridge vs. diagonalized. Same-data comparison (50 replicates, $\lambda = 0.3$): the full ridge estimator is uniformly better (median full/diag risk ratio $0.64/0.50$ on spec-

tral families A/B), *i.e.*, the feedthrough "noise" is partly recoverable signal for the full estimator; on the Σ_r-weighted risk our formula overestimates by 8% (A) / 28% (B). The phase structure (sign conditions, transitions) is unchanged; quantitative thresholds shift, motivating the T1-b conjecture below.

(v9) Falsification of a naive assembly. A leading-order power-law assembly predicts $n_r^{\text{crit}} \propto \mu^{-2.05}$; the measured exponent is -1.076. Diagnosis: the assembly drops the harm channel beyond the μ-induced truncation $j_\mu = \mu^{-1/\alpha_r}$ (where the cross term is amplified by μ^{-2}) and the mismatch-induced inflation of the feedthrough energy F_h. This failure is what led us to the H-functional form; we report it as the worked example of the falsification loop.

(v8) Harmful-region phase diagrams. Spectral family A, $n_r \in [10^2, 10^5]$, $n_h/n_r \in [0.5, 100]$: harmful-cell counts in Table 1 of the main text; the boundary expands monotonically in $\|\delta\|$ and toward small n_r.

(v12) Exchange-rate numerics as in Sec. G.

(v13) Unimodality verification. Randomized checks of the results of Sec. G (`verify_kex_unimodality.py`): the affine derivative identity of Lemma 6 matches finite differences to 7×10^{-6} (the finite-difference accuracy) over 2×10^5 random draws; the forbidden sign pattern of Proposition 7 never occurs, and all $\sim 10^5$ interior roots found are minima; the quadratic form of Theorem 8 holds to 2×10^{-14} with the closed-form λ^* matching the numerical argmin; condition (22) implies convexity in every one of the 1.8×10^4 draws in which it holds; and a targeted search over 4×10^3 aggregate risks (power-law and adversarial spectra) found no multimodal instance.

(v14) Dosage law and closed-form κ_{ex}. (`verify_dosage_law.py`.) Corollary 10: in 3.96×10^4 random general-spectrum draws with both endpoint signs strict, the global minimizer is interior in 100% of cases. Theorem 12: over 212 matched-spectrum configurations (8 decades of n_h each), the closed-form gain, the closed-form κ_{ex}, and the n_h^ε inversion match the per-coordinate ground truth (pointwise evaluation / bisection of the defining equation) to 3×10^{-10}, 1.1×10^{-10}, and 3×10^{-16} respectively, and $\lambda^*(n_h)$ is nondecreasing in every tested configuration.

(v16) Full-ridge closed form. (`verify_full_ridge.py`.) Monte-Carlo risk of the full ridge estimator against Eq. (27): relative errors between -8.4% and $+0.4\%$ across four configurations, decaying with sample size ($-2.5\% \to +0.3\%$ as n_r grows $100 \to 6400$); on the same configurations the T1-a form overestimates by $+4\%$ to $+61\%$, confirming the feedthrough-recovery prediction of consequence (ii); the mismatch sandwich converges to its per-coordinate price (errors -0.3% to $+8\%$).

I. A label-shift floor for aligned representations (boundary)

This section formalizes boundary (a) stated in the main text: when alignment is enforced on representations whose *action marginals* differ, a shared deterministic head provably fails on the target domain. It is the squared-loss, mixture-distribution analogue of the classical bound of Zhao et al. (cited in the main text).

Setup: a representation $z = \Phi(x)$ with induced feature laws P_r^z, P_h^z satisfying $W_1(P_r^z, P_h^z) \le \varepsilon$; action marginals P_e^a with means μ_e, gap $\Delta\mu = \mu_h - \mu_r$; a deterministic L-Lipschitz head f; domain risks $R_e(f) = \mathbb{E}[(f(z) - y_e)^2]$.

Proposition 13 (label-shift floor)**.** *For every deterministic head f:*

(i) $R_r(f) + R_h(f) \ge \frac{1}{2}\big(|\Delta\mu| - L\varepsilon\big)_+^2$;

(ii) the mixture risk $R_\lambda(f) = (1-\lambda)R_r(f) + \lambda R_h(f)$ obeys

$$R_\lambda(f) \ge \lambda(1-\lambda)\big(|\Delta\mu| - L\varepsilon\big)_+^2; \tag{26}$$

(iii) any head with $R_\lambda(f) \le \eta$ pays target risk $R_r(f) \ge \big(|\Delta\mu| - L\varepsilon - \sqrt{\eta/\lambda}\big)_+^2$; in particular the alignment-constrained mixture-optimal head pays $R_r \ge \lambda^2\big(|\Delta\mu| - L\varepsilon\big)_+^2$: the target floor grows quadratically *in the mixing ratio.*

Proof. Write $\nu_e = \mathbb{E}_{z \sim P_e^z}[f(z)]$. By Jensen, $R_e(f) \ge (\nu_e - \mu_e)^2$, and by the Kantorovich–Rubinstein dual, $|\nu_r - \nu_h| \le L\, W_1(P_r^z, P_h^z) \le L\varepsilon$. (i) $(\nu_r - \mu_r)^2 + (\nu_h - \mu_h)^2 \ge \frac{1}{2}\big((\nu_r - \mu_r) - (\nu_h - \mu_h)\big)^2 \ge \frac{1}{2}(|\Delta\mu| - L\varepsilon)_+^2$, since $(\nu_r - \mu_r) - (\nu_h - \mu_h) = -\Delta\mu + (\nu_r - \nu_h)$. (ii) Minimize $(1-\lambda)(\nu_r - \mu_r)^2 + \lambda(\nu_h - \mu_h)^2$ subject to $|\nu_r - \nu_h| \le L\varepsilon$. For fixed shift $\delta = \nu_h - \nu_r$, the minimum over the common level is attained at $\nu_r = (1-\lambda)\mu_r + \lambda(\mu_h - \delta)$ with value $\lambda(1-\lambda)(\Delta\mu - \delta)^2$; minimizing over $\delta \in [-L\varepsilon, L\varepsilon]$ gives $\delta^* = \operatorname{sign}(\Delta\mu)\min(L\varepsilon, |\Delta\mu|)$ and value $\lambda(1-\lambda)(|\Delta\mu| - L\varepsilon)_+^2$. (iii) $R_\lambda(f) \le \eta$ implies $|\nu_h - \mu_h| \le \sqrt{\eta/\lambda}$, hence $|\nu_r - \mu_r| \ge |\Delta\mu| - L\varepsilon - \sqrt{\eta/\lambda}$. For the constrained mixture optimum, $\nu_r - \mu_r = \lambda(\Delta\mu - \delta^*)$, so $R_r \ge \lambda^2(|\Delta\mu| - L\varepsilon)_+^2$. □

Remark 4 (Bernoulli case). For binary actions $y \in \{0, 1\}$, $|\Delta\mu| = D_{\text{TV}}(P_r^a, P_h^a)$ and (i) is the Brier-score analogue of the classification bound of Zhao et al.: perfectly aligned representations with shifted label marginals are guaranteed to fail — exact alignment ($\varepsilon \to 0$) does not remove the floor, it creates it. For continuous actions the first-moment gap is the sharp quantity; second-moment (variance) shifts give analogous floors with second moments in place of means.

Remark 5 (operational reading for the L1 test). The floor is monotone in the marginal gap and, for the mixture-optimal head, quadratic in λ. This is the quantitative prediction for

the action-marginal-shift axis of the L1 grid (main text, Experiments): inject a controlled marginal shift into the human action labels, enforce alignment, and the target-domain success should hit a floor that moves with λ^2. Numerical checks (`verify_zhao_floor.py`): bounds (i)–(iii) hold on 2×10^5 random draws with zero violations, and the constrained-optimum closed form matches brute-force minimization.

J. Theorem T1-b: the full-ridge deterministic equivalent (fixed-p regime)

We now upgrade the conjecture announced in earlier drafts to a theorem in the fixed-dimension regime. The full ridge estimator is $\hat{\beta} = A^{-1}(w_r X_r^\top y_r + w_h X_h^\top y_h)$ with $A = w_r S_r + w_h S_h + \mu I$ and $S_e = X_e^\top X_e$ the (non-diagonal) sample Gram matrices, in the model of Sec. A.

Theorem 14 (T1-b). *Let the covariates be Gaussian, p fixed, $\mu > 0$ fixed, and $n_r, n_h \to \infty$ at fixed $\lambda \in (0,1)$. Then*

$$\mathcal{R}_T(\lambda) = \sum_j \frac{a_j}{D_j^2}\Big[V_j(\lambda) + \big(\lambda b_j \delta_j - \mu \beta_j^*\big)^2\Big] + O_p\big(n^{-1/2}\big), \tag{27}$$

with $V_j(\lambda) = (1-\lambda)^2 \sigma_r^2\, a_j/n_r + \lambda^2 \sigma_h^2\, b_j/n_h$, $D_j = (1-\lambda)a_j + \lambda b_j + \mu$, and $n = \min(n_r, n_h)$; the O_p constants depend on $(p, \mu^{-1}, \|\beta^*\|, \|\delta\|)$ and the spectra. In words: the full-ridge deterministic equivalent is the T1-a closed form with the feedthrough-corrected noise $\tilde{\sigma}_e^2 = \sigma_e^2 + F_e$ replaced by the true label noise σ_e^2.

Proof. The error decomposes exactly as

$$\hat{\beta} - \beta^* = -\mu A^{-1}\beta^* + A^{-1} w_h S_h\, \delta + A^{-1}\eta, \qquad \eta = w_r X_r^\top \varepsilon_r + w_h X_h^\top \varepsilon_h. \tag{28}$$

Two estimates drive everything: (1) Wishart concentration, $\|S_e/n_e - \Sigma_e\| = O_p(\sqrt{p/n_e})$, hence $\|A - D\| = O_p(n^{-1/2})$ with $D = (1-\lambda)\Sigma_r + \lambda\Sigma_h + \mu I = \mathrm{diag}(D_j)$; (2) resolvent perturbation, using the deterministic bound $\|A^{-1}\| \le 1/\mu$: on the event $\|A - D\| \le \mu/2$ (whose complement is exponentially unlikely),

$$\|A^{-1} - D^{-1}\| \le \|A - D\| \,/\, \big(\mu\,(\mu - \|A - D\|)\big) = O_p(n^{-1/2}). \tag{29}$$

For the mismatch term write $w_h S_h \delta = \lambda \Sigma_h \delta + \zeta$ with $\mathbb{E}\|\zeta\|^2 = (\lambda^2/n_h)\,\mathrm{Var}\big(x(x^\top\delta)\big) = O(1/n_h)$; hence the mismatch-plus-shrinkage risk converges to $\sum_j a_j(\lambda b_j \delta_j - \mu\beta_j^*)^2/D_j^2$, including the alignment cross term. For the variance, $\mathbb{E}[\eta\eta^\top \mid X] = w_r^2\sigma_r^2 S_r + w_h^2 \sigma_h^2 S_h$, so $\mathbb{E}[\eta^\top A^{-1}\Sigma_r A^{-1}\eta]$ converges to $\mathrm{tr}\,\Sigma_r D^{-1}\big((1-\lambda)^2\sigma_r^2\Sigma_r/n_r + \lambda^2\sigma_h^2\Sigma_h/n_h\big)D^{-1}$, the displayed variance sum. Cross terms involving η vanish by conditional centering, and all replacements of A^{-1} by D^{-1} are justified by the deterministic bound $\|\Sigma_r^{1/2} A^{-1} \Sigma_e^{1/2}\| \le (\max_j a_j \cdot \max_j \lambda_j^e)^{1/2}/\mu^2$ on the good event. □

Three consequences. (i) *The mismatch sandwich is confirmed*: $\mathcal{B}_{\mathrm{mis}} = w_h^2\, \delta^\top S_h A^{-1}\Sigma_r A^{-1} S_h\, \delta = \sum_j a_j \big(\lambda b_j \delta_j / D_j\big)^2 (1 + o(1))$ — the conjectured per-coordinate price, with $D_j^{\mathrm{full}} = D_j$ in this regime. (ii) *Feedthrough recovery*: the T1-a risk exceeds the full-ridge risk by $\sum_j a_j\big[(1-\lambda)^2 F_r a_j/n_r + \lambda^2 F_h b_j/n_h\big]/D_j^2$ to leading order — exactly the off-diagonal signal energy that the coordinate-wise estimator treats as noise. This explains the direction and magnitude of the v6 gap (full ridge uniformly better; 8–28% on the weighted risk), and upgrades "Eq. (T1-a) is a variance upper bound" from an observation to a formula. (iii) *All downstream results transfer*: Corollary 1, Theorem 2, and Corollary 2 of the main text, and the dosage law of Theorem 12, hold for the full ridge estimator with the substitution $\tilde{\sigma}_e^2 \to \sigma_e^2$; the phase structure is unchanged in form, only the variance prices shrink.

Remark 6 (the proportional regime). For $p \asymp n$ the resolvent perturbation step fails and an anisotropic deterministic equivalent is needed. The structure of a prospective proof is visible in the leave-one-out expansion of the sandwich: writing $S_h = \sum_i x_i x_i^\top$, the cross pairs $i \ne k$ carry the leading order through $\mathbb{E}[x\,(x^\top\delta)] = \Sigma_h\delta \ne 0$, while the diagonal pairs $i = k$ contribute only $O(1/n_h)$ corrections; a proportional-regime proof must therefore control $\sum_{i\ne k}(\delta^\top x_i)(x_i^\top A^{-1}\Sigma_r A^{-1} x_k)(\delta^\top x_k)$ via leave-two-out resolvent expansions — the route through Wu's pipeline flagged in the project notes. The low-rank fallback of earlier drafts is no longer needed in the fixed-p regime.

K. Experimental details

Training. CNN diffusion policy [1]; image observations (96×96), 2 observation steps / 8 action steps; Adam; 50 epochs; the mixture dataset samples human-domain sequences with probability λ per batch; the validation split is always robot-domain. **Evaluation.** 20 closed-loop episodes per run in the robot-domain simulator (episode seeds 101–120, horizon 300), coverage = mean final T-coverage (the binary success threshold 0.855 is never reached at this training budget); open-loop loss on held-out robot-domain sequences. **Probes.** Latent W_2: closed-form Gaussian (Fréchet) approximation on encoder features; domain AUC: 5-fold linear probe (saturated at $\approx$ 1.0 in batch 1 — a capacity-limited probe replaces it in batch 2); gradient cosine and $\kappa(\hat{G}) = 1/\sqrt{1-\cos^2}$ from per-domain gradients at the shared encoder (exact two-task form). **Data.** 122 robot-domain and 128 human-domain demonstrations (14.5k/15.5k steps) replayed from the public Push-T expert data; 102 paired episodes share initial configurations; mirror convention $\theta \mapsto -\theta$. **Compute.** 8×RTX 4090, one run per GPU (∼14.5 GB each), co-tenanted; the 24-run batch takes ≈1.4 h wall-clock.

References

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